

Pinker Charts - Recreated Figure Comparison Review

Single continuous-scroll review PDF. Each figure page uses the most complete available side-by-side comparison image found in the repository/worktrees and compares the recreated chart against the original reference crop from the supplemental graphics PDF. Source-recovery/status-only figures without real recreated charts are excluded.

Figure	Status	Selected comparison	Source branch/worktree
4-1	manual_review_needed	figure_4_1_extended_status_comparison.png	base_track_d_current
5-1	verified_reproduction	figure_5_1_extended_comparison.png	base_track_d_current
5-2	partial_match	figure_5_2_extended_comparison.png	base_track_d_current
5-3	partial_match	figure_5_3_extended_comparison.png	base_track_d_current
5-4	needs_targeted_source_recovery	figure_5_4_extended_comparison.png	base_track_d_current
6-1	blocked_external_source	figure_6_1_extended_comparison.png	base_track_d_current
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9-6	verified_reproduction	figure_9_6_extended_comparison.png	base_track_d_current
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12-9	verified_reproduction	figure_12_9_extended_comparison.png	base_track_d_current
19-1	partial_match	figure_19_1_extended_comparison.png	base_track_d_current

1. Figure 4-1: Tone of the news, 1945-2010

Status: manual_review_needed | Selected: figure_4_1_extended_status_comparison.png | Source: base_track_d_current

Deeper source recovery completed; no reconstruction because original monthly data remain unresolved. Status panels are not reconstructions.

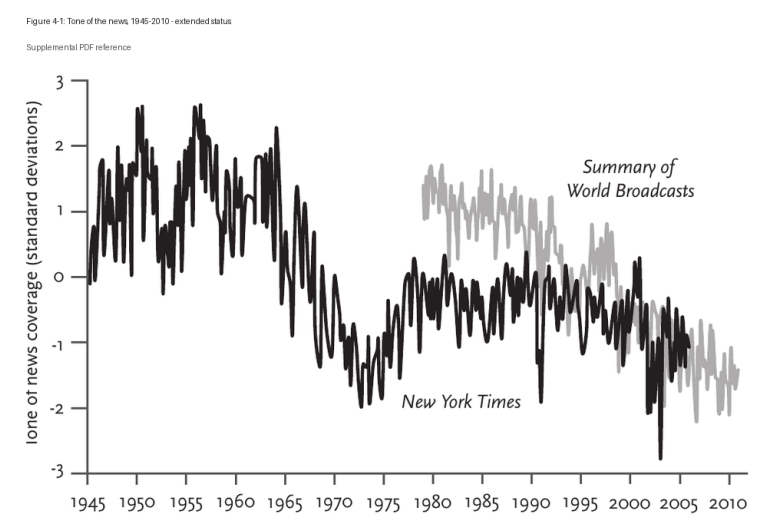


Figure 4-1: Tone of the news, 1945–2010
Source: Leetaru 2011. Plotted by month, beginning in January.

Source-recovery status panel

Figure 4-1: Tone of the news, 1945-2010

Source recovery blocked

No accepted underlying monthly data table was recovered.
Leetaru/GDELT high-resolution PNGs were preserved as source-chain evidence only.
The book/supplemental chart and Leetaru plot images were not digitized as data.
Required next step: recover the original monthly NYT and SWB tone series,
or a reproducible extraction from the original text corpora and sentiment method.
No extension is plotted until the book-period source is recovered.

Status: manual_review_needed | Visual artifact is a status panel, not a reconstruction.

2. Figure 5-1: Life expectancy, 1771-2015

Status: verified_reproduction | Selected: figure_5_1_extended_comparison.png | Source: base_track_d_current

Book-period reconstruction is verified. The extended artifact intentionally plots no successor segment because a comparable regional successor series has not been accepted.

Figure 5-1 extended comparison

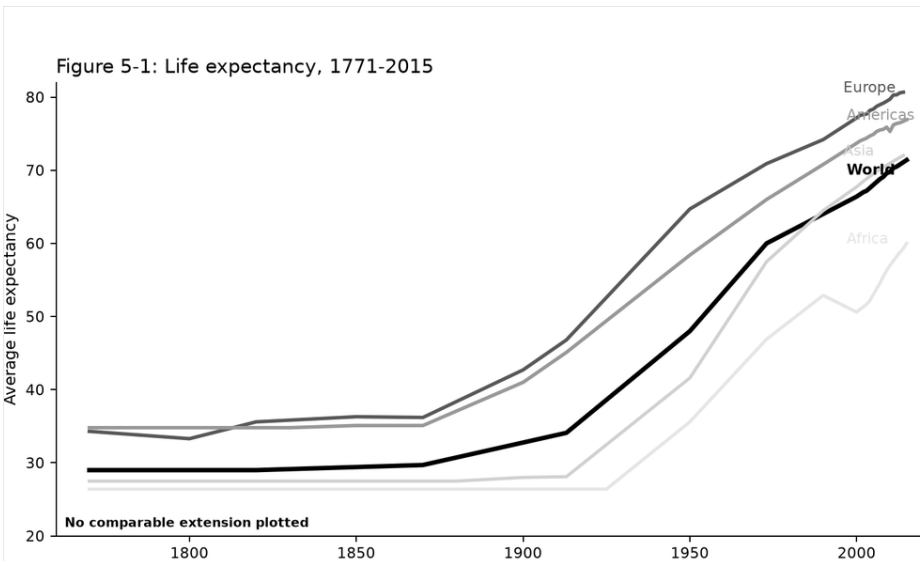
PDF chart reference



Figure 5-1: Life expectancy, 1771–2015

Sources: *Our World in Data*, Roser 2016n, based on data from Riley 2005 for the years before 2000 and from the World Health Organization and the World Bank for the subsequent years. Updated with data provided by Max Roser.

Recreated



Source: OWID/Roser 2016n historical dataset. No comparable regional extension plotted; see caption.

3. Figure 5-2: Child mortality, 1751-2013

Status: partial_match | Selected: figure_5_2_extended_comparison.png | Source: base_track_d_current

Remediated to use the current OWID selected child-mortality grapher directly in percent units. This materially improves the visual match, but the exact Roser 2016a UN/HMD vintage remains unrecovered.

Figure 5-2 extended comparison

PDF chart reference

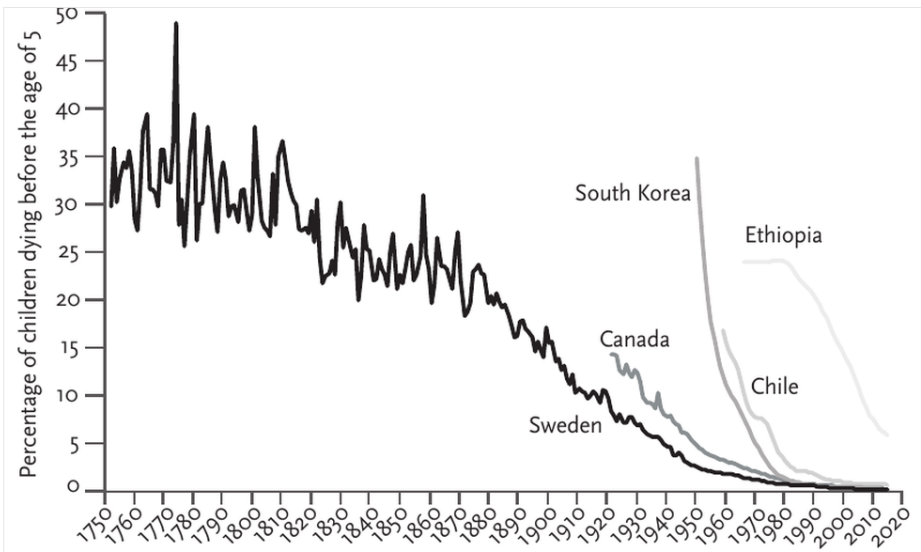
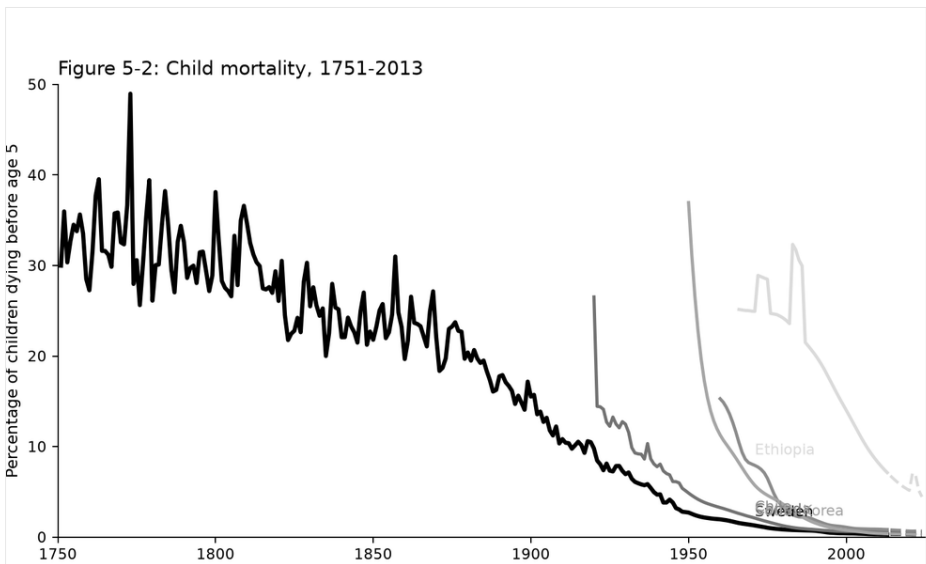


Figure 5-2: Child mortality, 1751–2013

Sources: Our World in Data, Roser 2016a, based on data from the UN Child Mortality estimates, <http://www.childmortality.org/>, and the Human Mortality Database, <http://www.mortality.org/>.

Recreated



Improved proxy: current OWID selected child-mortality grapher; exact Roser 2016a vintage unrecovered.

4. Figure 5-3: Maternal mortality, 1751-2013

Status: partial_match | Selected: figure_5_3_extended_comparison.png | Source: base_track_d_current

Track A health/nutrition batch. Source status: updated_equivalent.

Figure 5-3 extended comparison

PDF chart reference

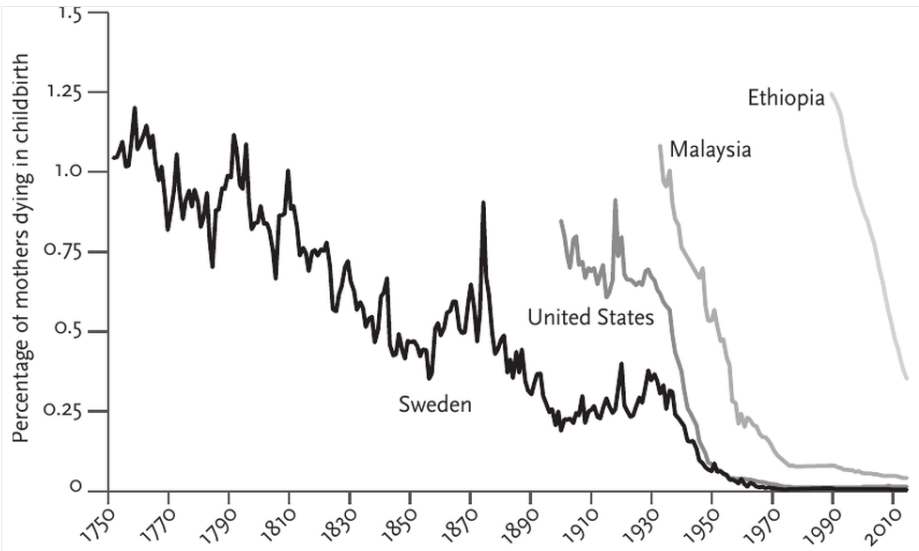
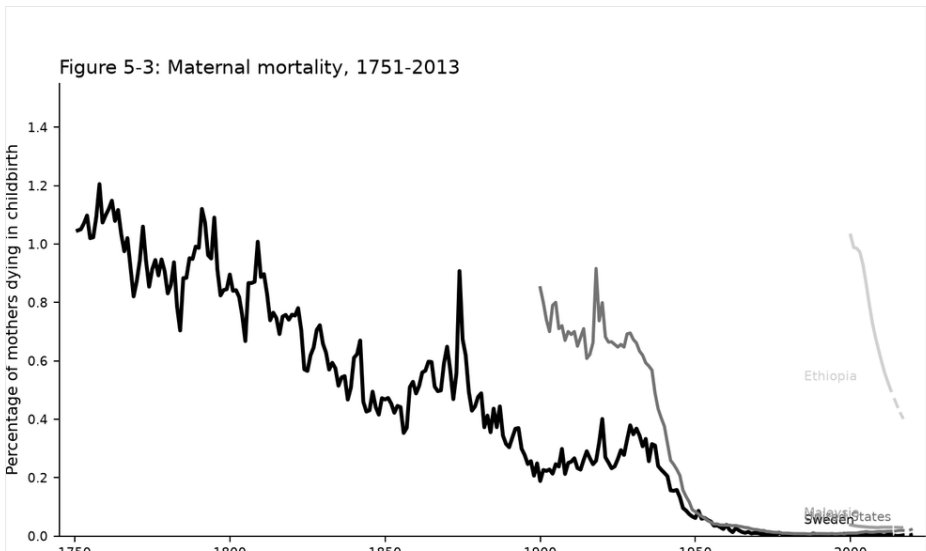


Figure 5-3: Maternal mortality, 1751–2013

Source: Our World in Data, Roser 2016p, based partly on data from Claudia Hanson of Gapminder, <https://www.gapminder.org/data/documentation/gdo10/>.

Recreated



Proxy: current OWID maternal-mortality ratio converted from per 100,000 births to percent.

5. Figure 5-4: Life expectancy, UK, 1701-2013

Status: needs_targeted_source_recovery | Selected: figure_5_4_extended_comparison.png | Source: base_track_d_current

Track A health/nutrition batch. Source status: needs_targeted_source_recovery.

Figure 5-4 extended comparison

PDF chart reference

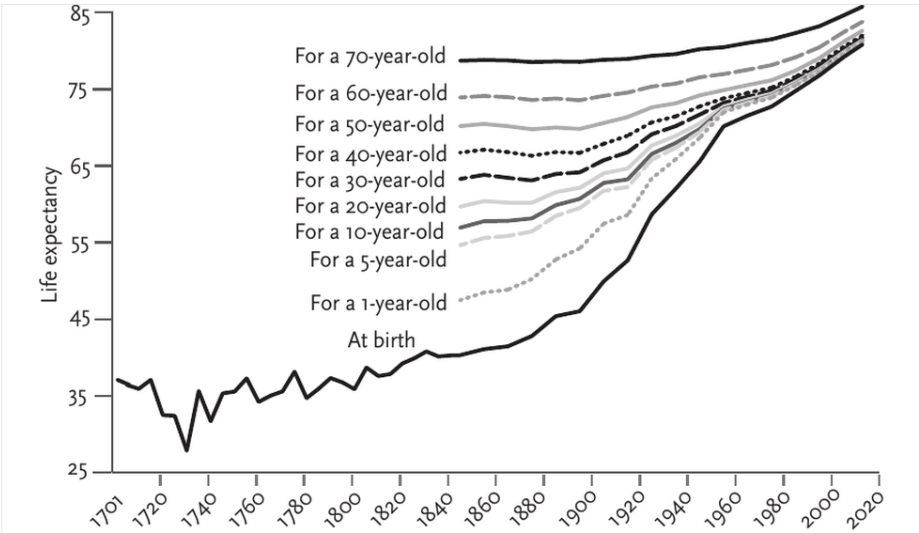
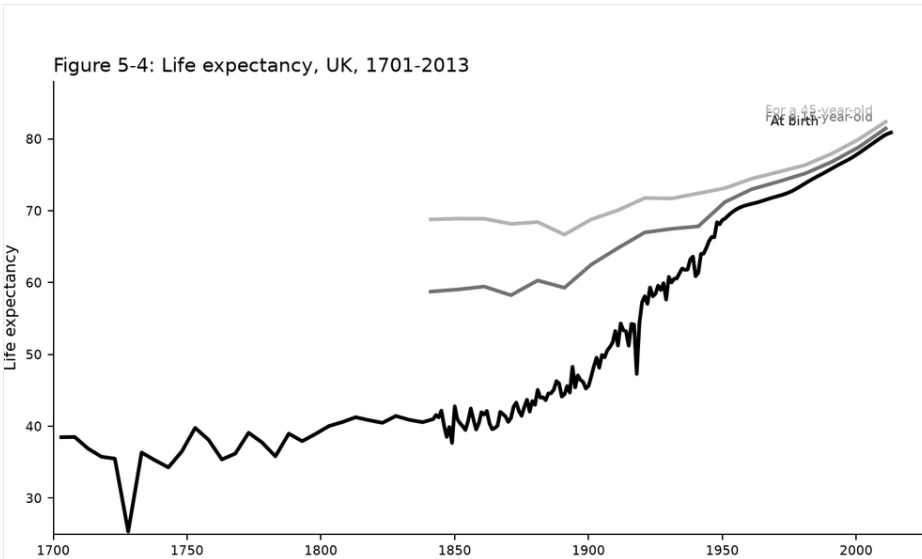


Figure 5-4: Life expectancy, UK, 1701-2013

Sources: *Our World in Data*, Roser 2016n. Data before 1845 are for England and Wales and come from OECD Clio Infra, van Zanden et al. 2014. Data from 1845 on are for mid-decade years only, and come from the *Human Mortality Database*, <http://www.mortality.org/>.

Recreated



Partial reconstruction only: exact HMD age-1/5/10/20/30/40/50/60/70 series not recovered.

6. Figure 6-1: Childhood deaths from infectious disease, 2000-2013

Status: blocked_external_source | Selected: figure_6_1_extended_comparison.png | Source: base_track_d_current

Track A health/nutrition batch. Source status: blocked_external_source.

Figure 6-1 extended comparison

PDF chart reference

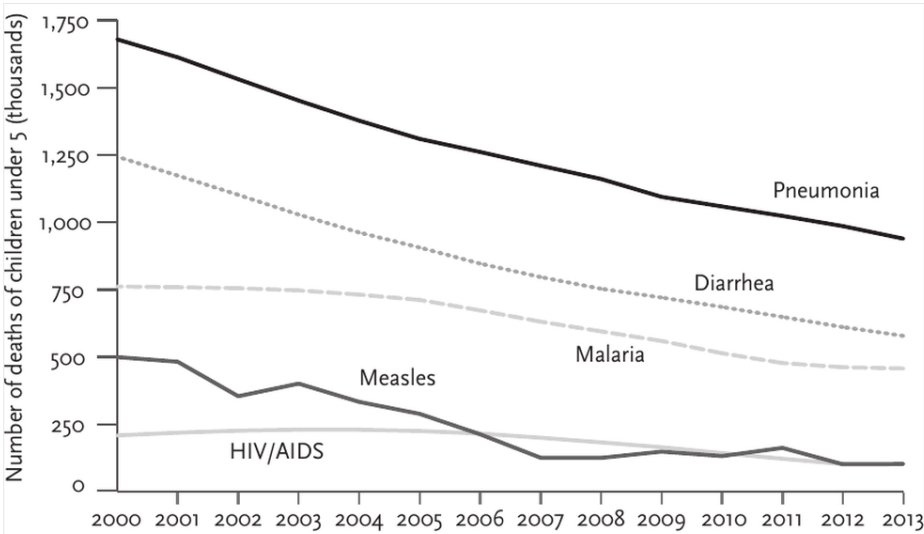
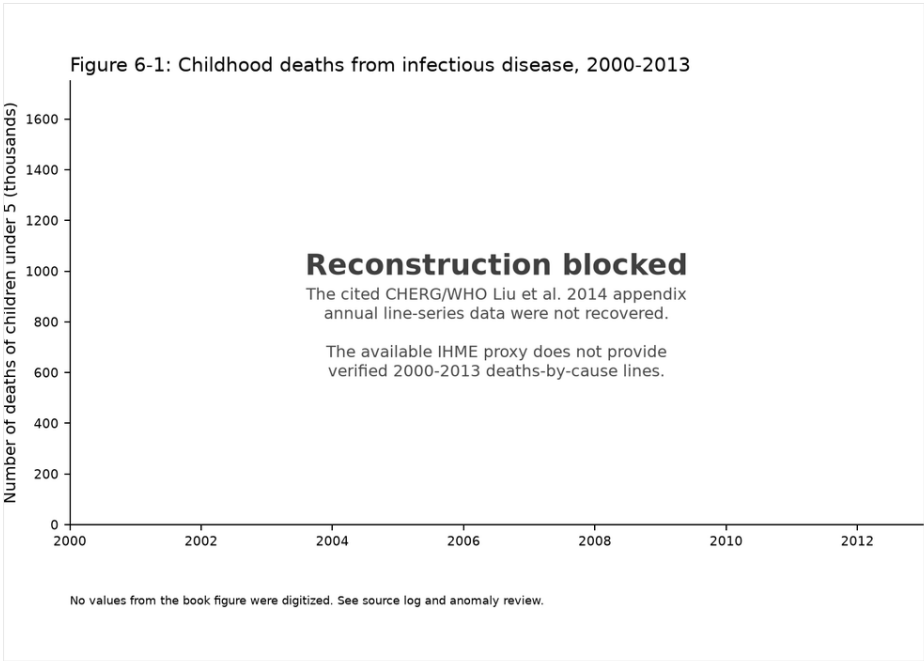


Figure 6-1: Childhood deaths from infectious disease, 2000–2013

Source: Child Health Epidemiology Reference Group of the World Health Organization, Liu et al. 2014, supplementary appendix.

Recreated



7. Figure 7-1: Calories, 1700-2013

Status: partial_match | Selected: figure_7_1_extended_comparison.png | Source: base_track_d_current

Track A health/nutrition batch. Source status: updated_equivalent.

Figure 7-1 extended comparison

PDF chart reference

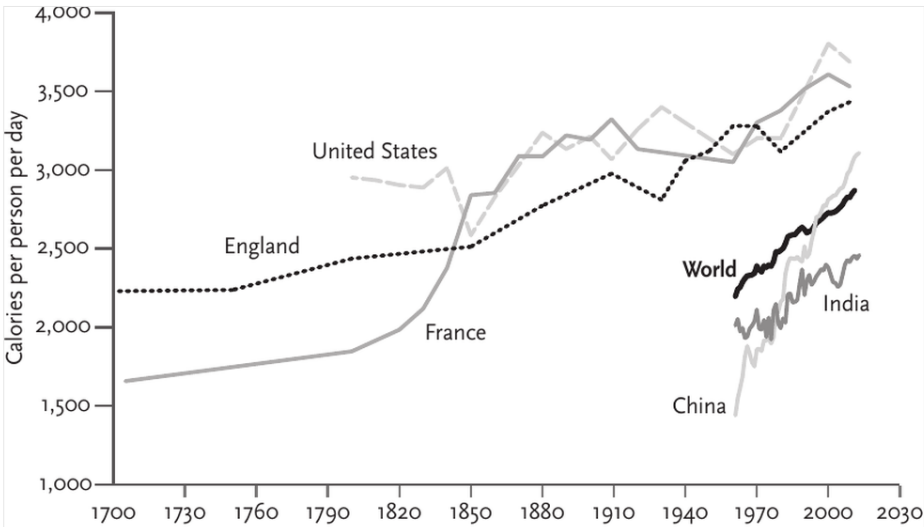
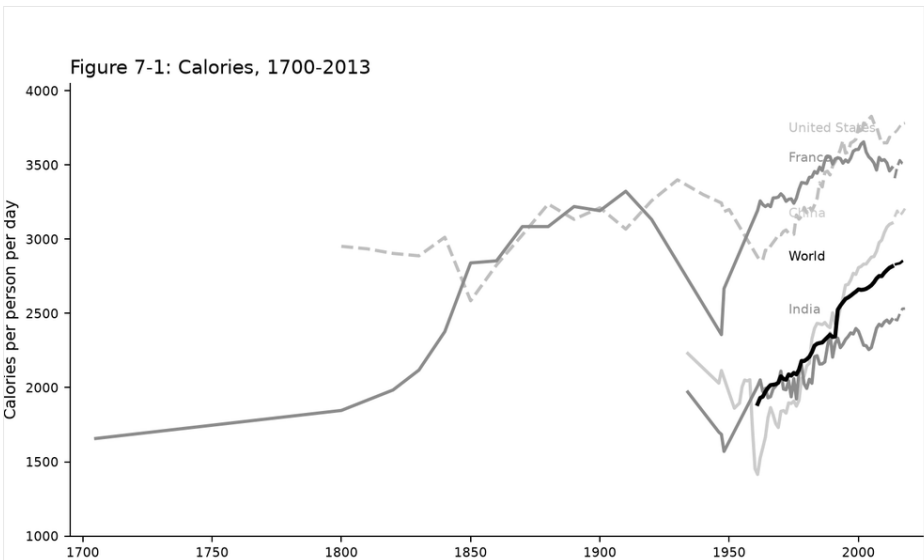


Figure 7-1: Calories, 1700–2013

Sources: United States, England, and France: *Our World in Data*, Roser 2016d, based on data from Fogel 2004. China, India, and the World: Food and Agriculture Organization of the United Nations, <http://www.fao.org/faostat/en/#data>.

Recreated



Source: OWID calories dataset based on FAO and historical sources; post-2013 values are successor continuation.

8. Figure 7-2: Childhood stunting, 1966-2014

Status: partial_match | Selected: figure_7_2_extended_comparison.png | Source: base_track_d_current

Track A health/nutrition batch. Source status: partial_match.

Figure 7-2 extended comparison

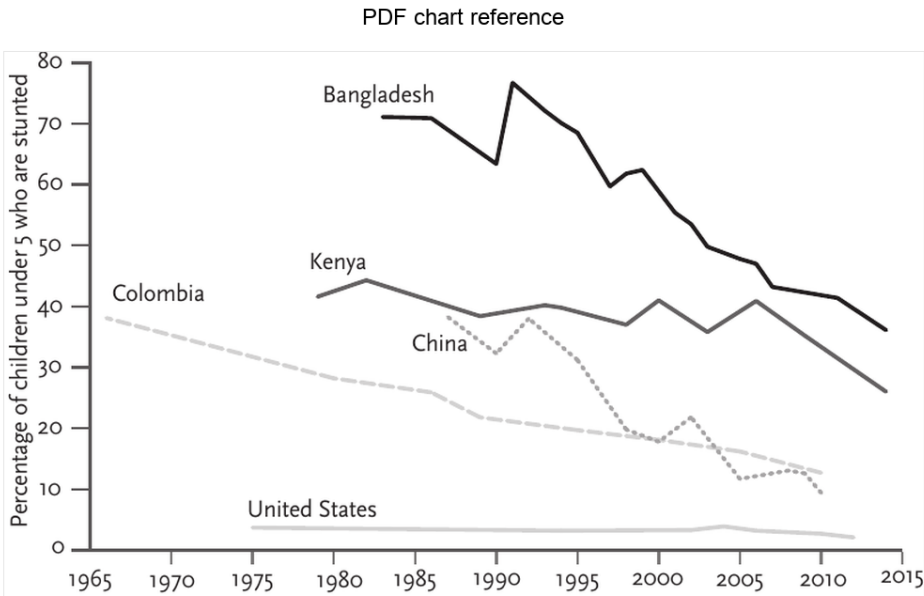
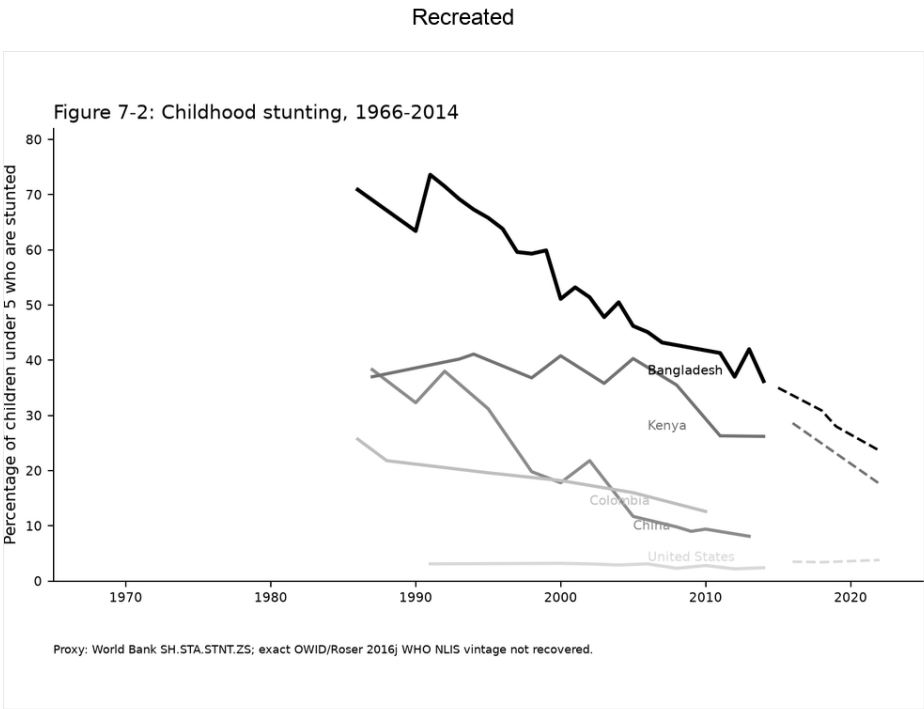


Figure 7-2: Childhood stunting, 1966–2014

Source: *Our World in Data*, Roser 2016j, based on data from the World Health Organization's *Nutrition Landscape Information System*, <http://www.who.int/nutrition/nlis/en/>.



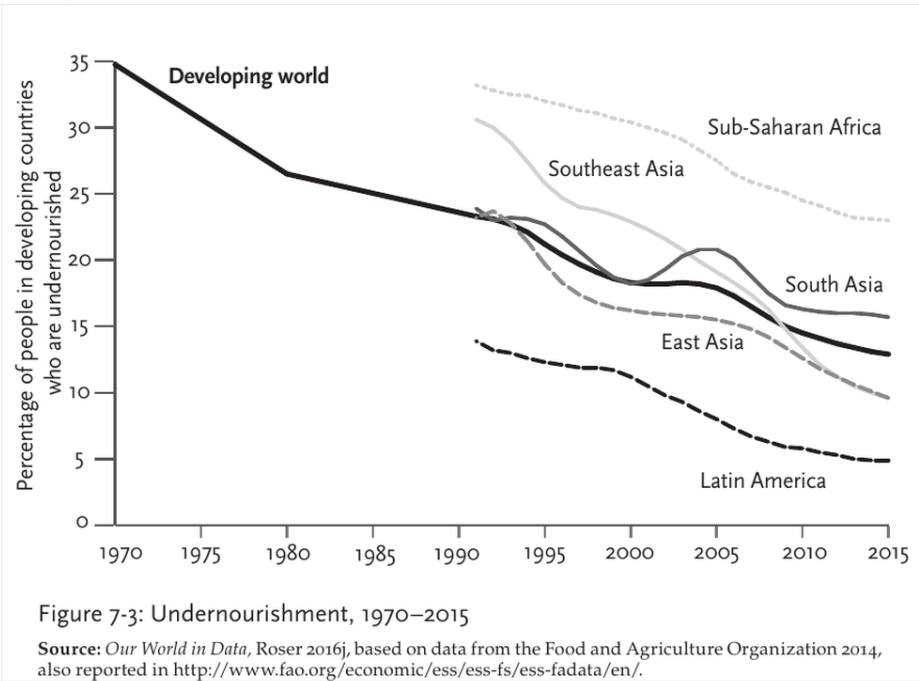
9. Figure 7-3: Undernourishment, 1970-2015

Status: partial_match | Selected: figure_7_3_extended_comparison.png | Source: base_track_d_current

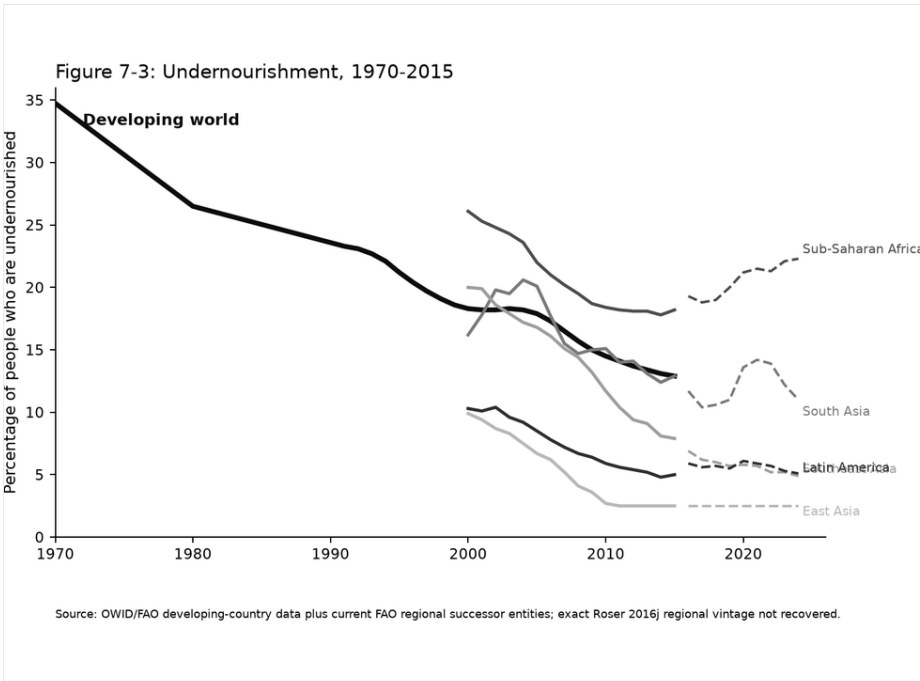
Fresh review found that the prior reconstruction substituted broad Africa/Asia/World curves for several book regional curves. The remediation replaces those with current FAO successor regional entities where available, but the exact Roser 2016j/FAO 2014 regional vintage and 1991-1999 regional segments remain unrecovered.

Figure 7-3 extended comparison

Book/PDF reference



Recreated

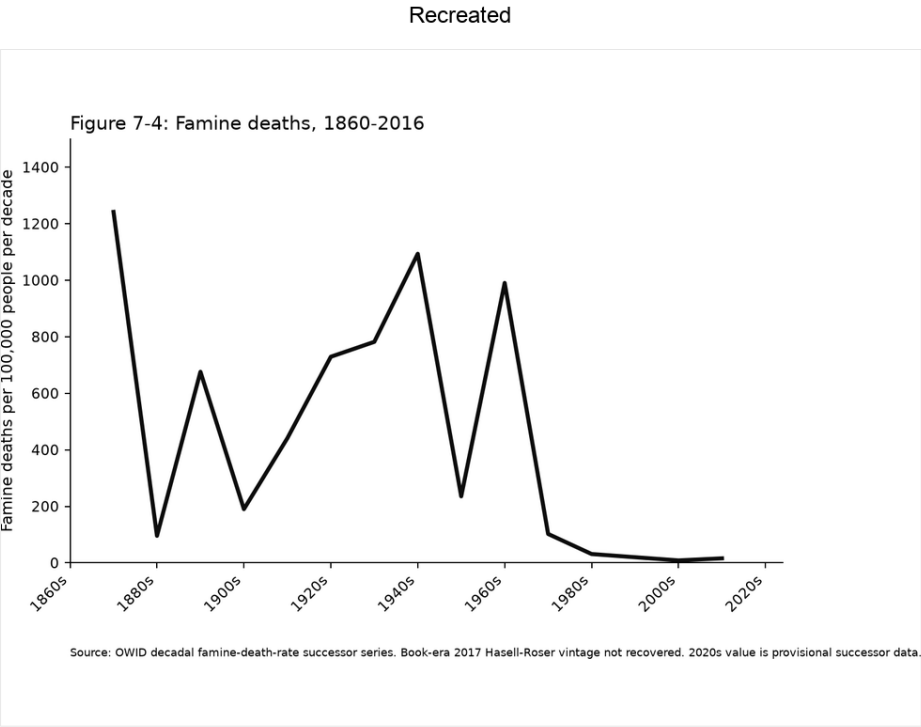
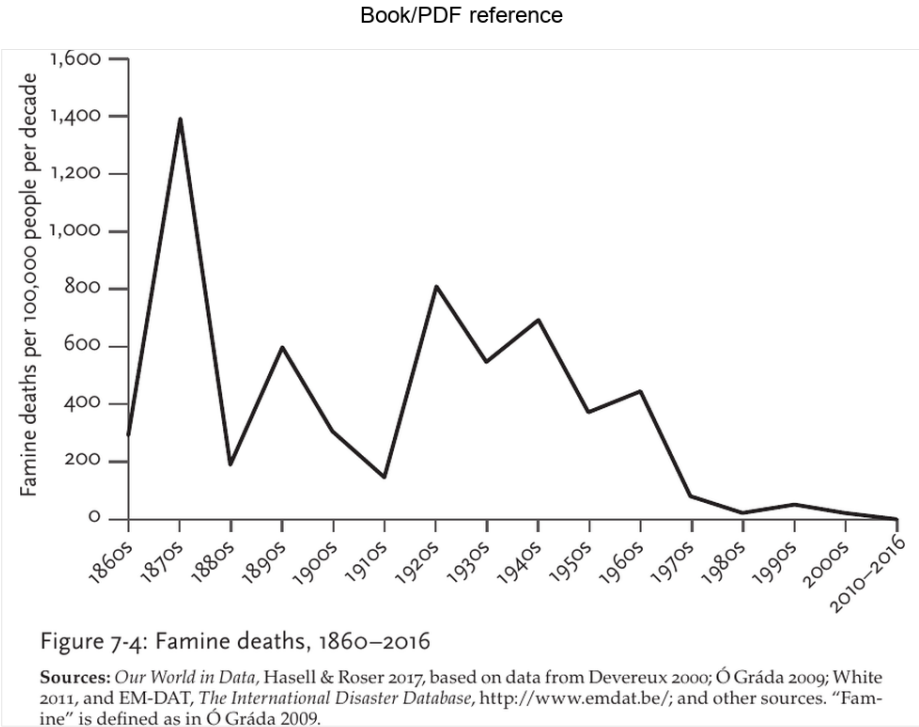


10. Figure 7-4: Famine deaths, 1860-2016

Status: partial_match | Selected: figure_7_4_extended_comparison.png | Source: base_track_d_current

The previous reconstruction recomputed rates from an event table with a current population denominator. Remediation uses OWID's explicit decadal famine-death-rate grapher series, which fixes the aggregation treatment, but it is a 2025 World Peace Foundation/OWID successor rather than the archived 2017 Hasell-Roser book vintage.

Figure 7-4 extended comparison



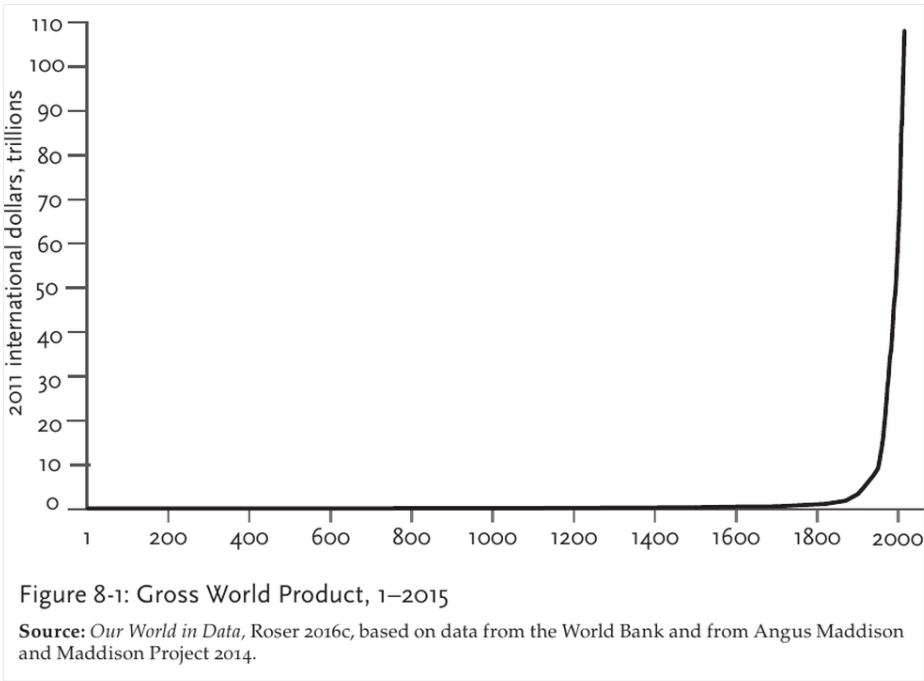
11. Figure 8-1: Gross World Product, 1-2015

Status: updated_equivalent | Selected: figure_8_1_extended_comparison.png | Source: base_track_d_current

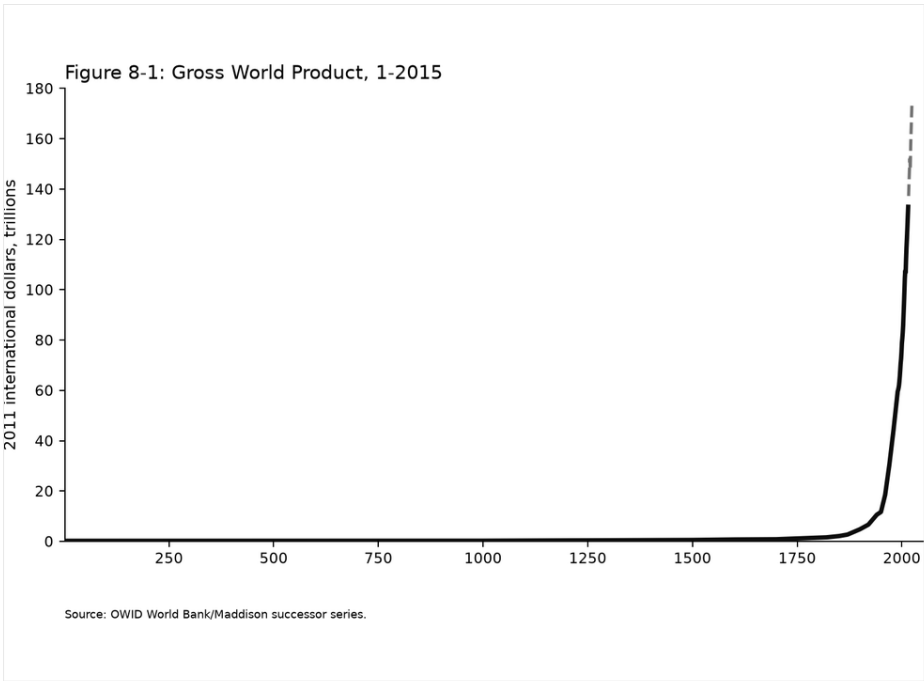
The successor OWID/Maddison/World Bank series preserves the hockey-stick shape and book-period coverage, but the 2015 scale is materially above the printed figure's apparent endpoint, so this is an updated equivalent rather than an exact reproduction.

Figure 8-1 extended comparison

Book/PDF reference



Recreated



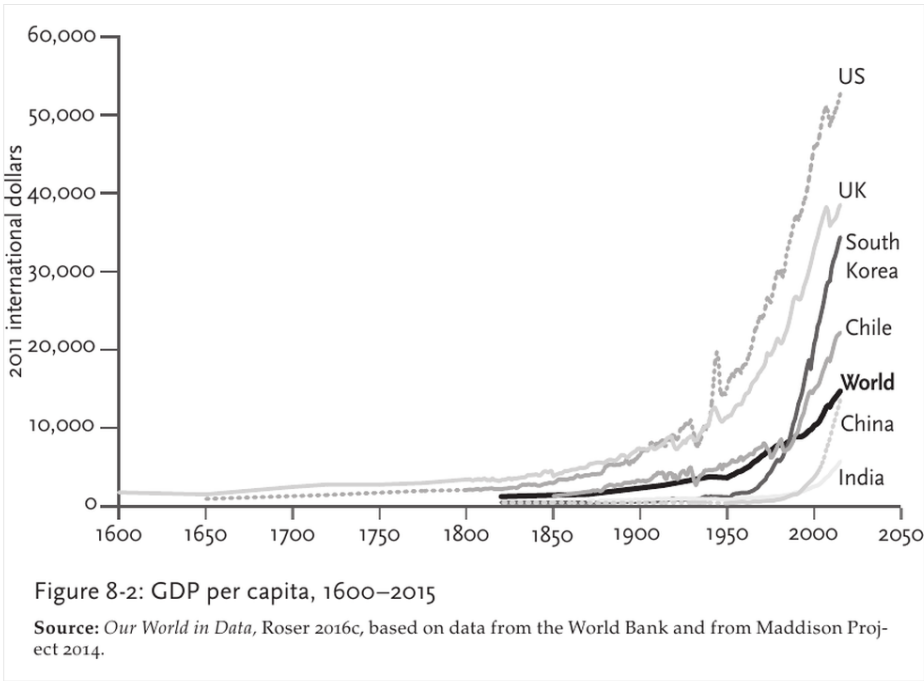
12. Figure 8-2: GDP per capita, 1600-2015

Status: updated_equivalent | Selected: figure_8_2_extended_comparison.png | Source: base_track_d_current

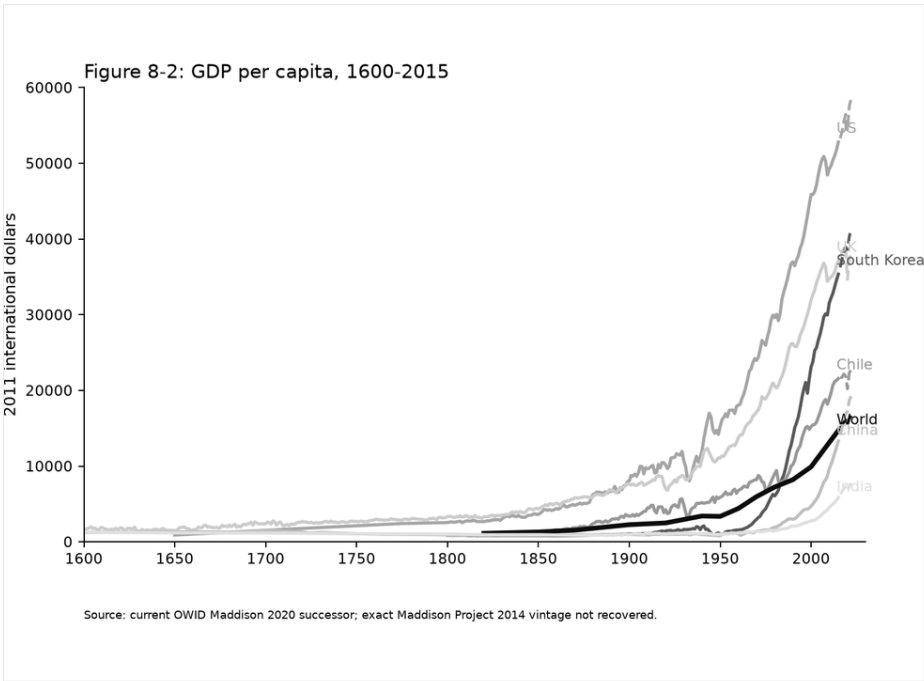
The selected countries and line ordering broadly match the book, but current Maddison 2020/OWID data are not the cited Maddison Project 2014/World Bank 2016 vintage. Remaining differences are not typography-only.

Figure 8-2 extended comparison

Book/PDF reference



Recreated

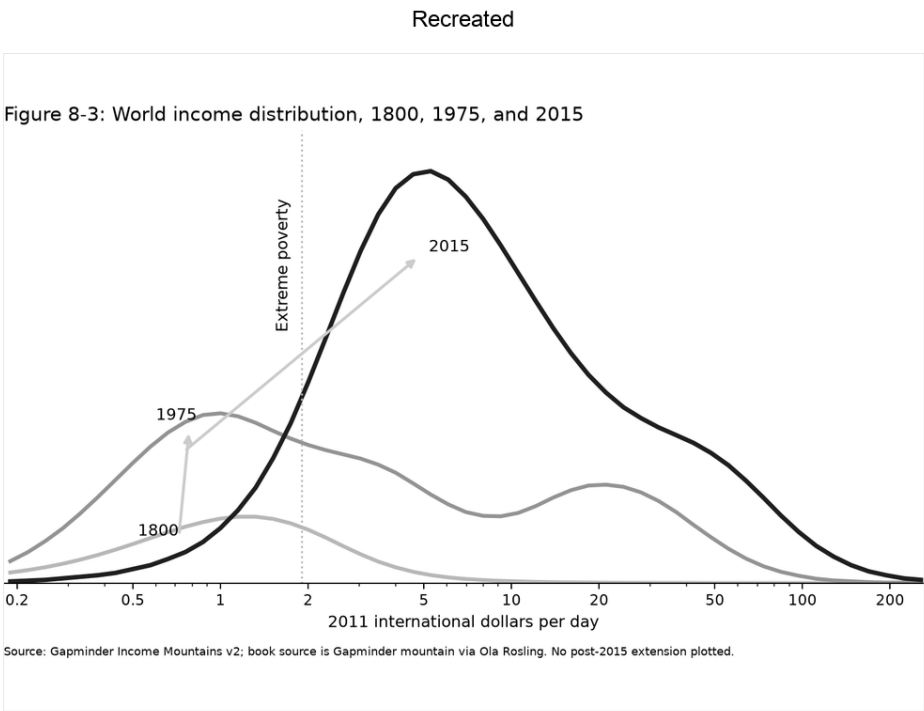
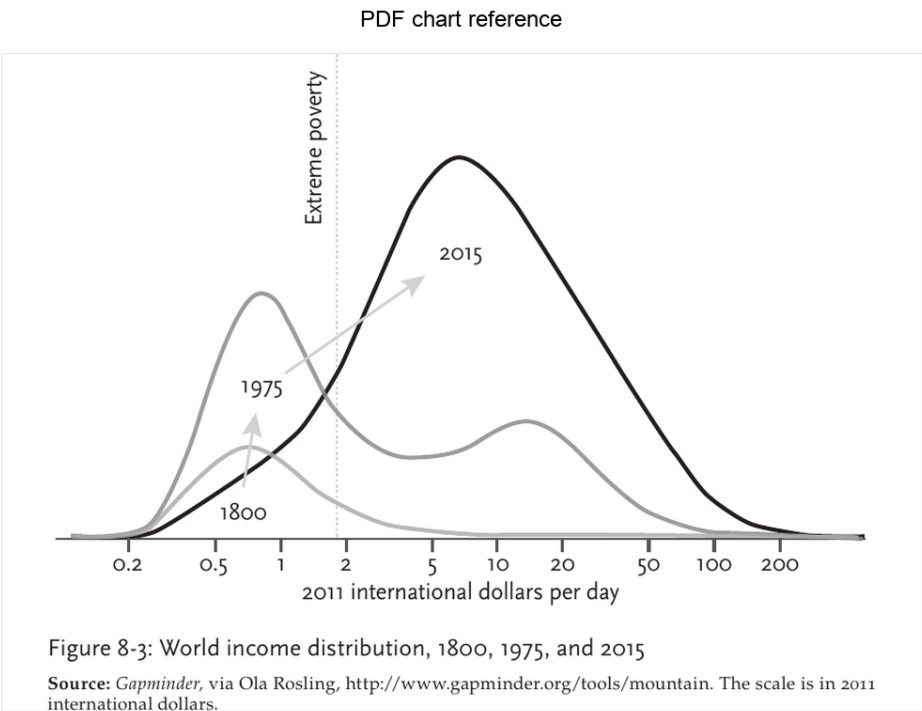


13. Figure 8-3: World income distribution, 1800, 1975, and 2015

Status: updated_equivalent | Selected: figure_8_3_extended_comparison.png | Source: base_track_d_current

Same Gapminder source family and variables were located, but the downloaded v2 workbook is a 2018 successor rather than a proven book-era snapshot. Shares were multiplied by workbook population to plot people in each income bin, matching the visible figure geometry more closely than normalized shares.

Figure 8-3 extended comparison



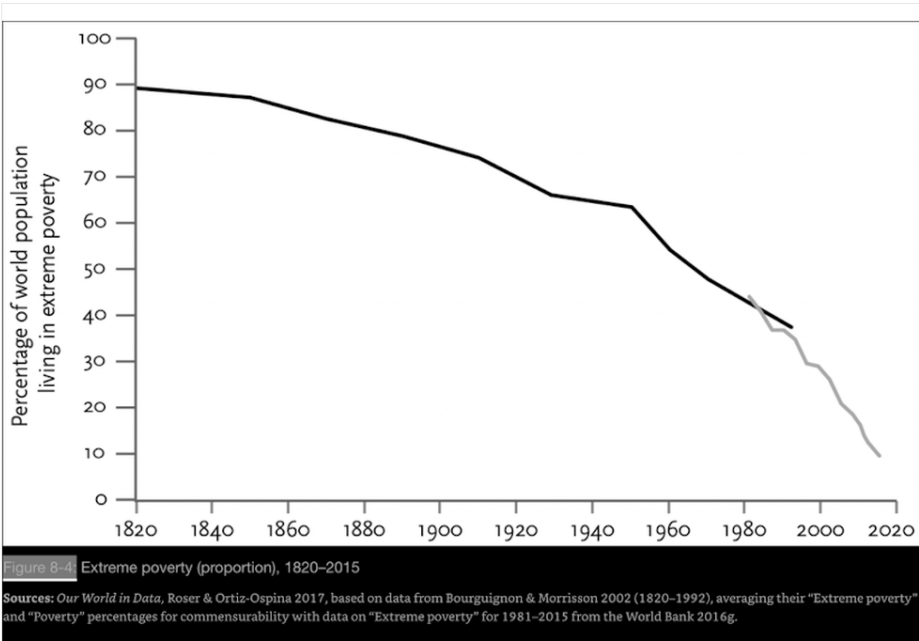
14. Figure 8-4: Extreme poverty (proportion), 1820-2015

Status: verified_reproduction | Selected: figure_8_4_extended_comparison.png | Source: base_track_d_current

Book-period reconstruction is verified. The comparison layout was remediated so the recreated plot area is comparable to the Kindle crop; no comparable world extension is plotted.

Figure 8-4 extended comparison

Kindle reference



Recreated



15. Figure 8-5: Extreme poverty (number), 1820-2015

Status: verified_reproduction | Selected: figure_8_5_extended_comparison.png | Source: base_track_d_current

A falsification audit found no material source, encoding, or endpoint issue. The reconstruction uses the cited OWID historical absolute-count series and matches the stacked-area visual claim; remaining differences are production styling only.

Figure 8-5 extended comparison

Book/PDF reference

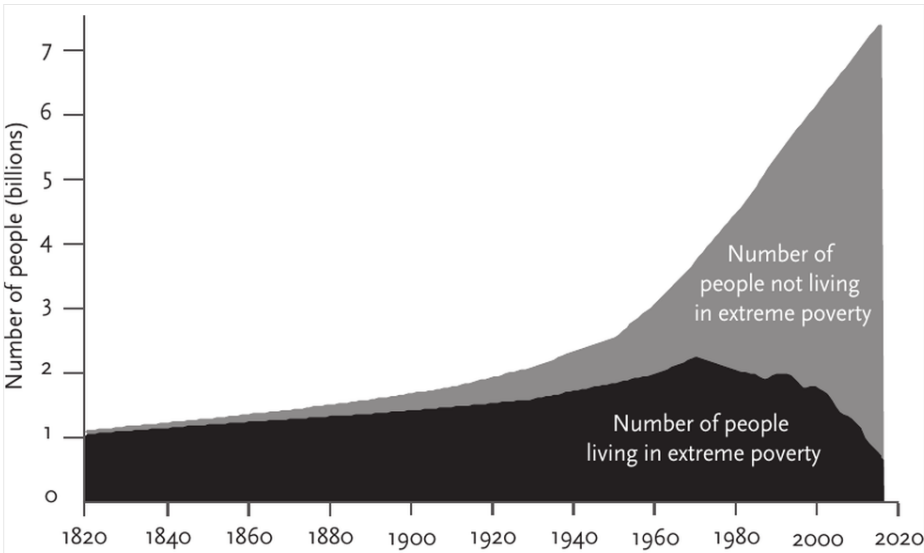
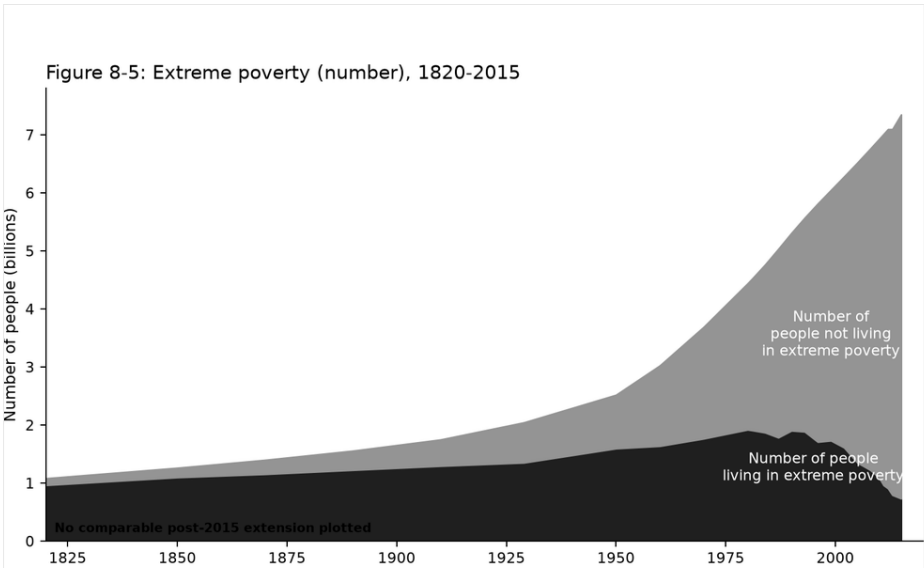


Figure 8-5: Extreme poverty (number), 1820–2015

Sources: *Our World in Data*, Roser & Ortiz-Ospina 2017, based on data from Bourguignon & Morrison 2002 (1820–1992) and the World Bank 2016g (1981–2015).

Recreated



Source: OWID Roser & Ortiz-Ospina historical poverty counts, Bourguignon & Morrison plus World Bank 2016g.

16. Figure 9-4: Social spending, OECD countries, 1880-2016

Status: updated_equivalent | Selected: figure_9_4_extended_comparison.png | Source: base_track_d_current

For verification, recover the exact OWID Ortiz-Ospina & Roser 2016b/2017 chart data snapshot; otherwise keep updated_equivalent status.

Figure 9-4 extended comparison

Book reference

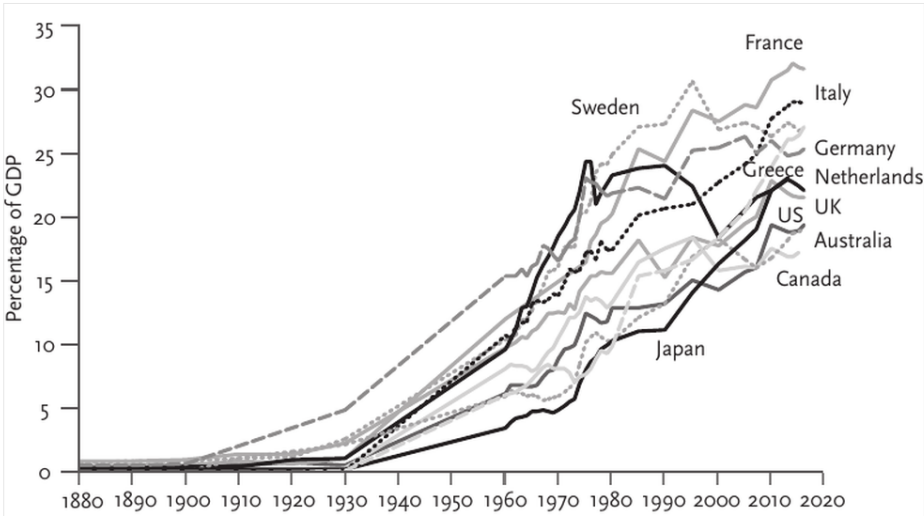
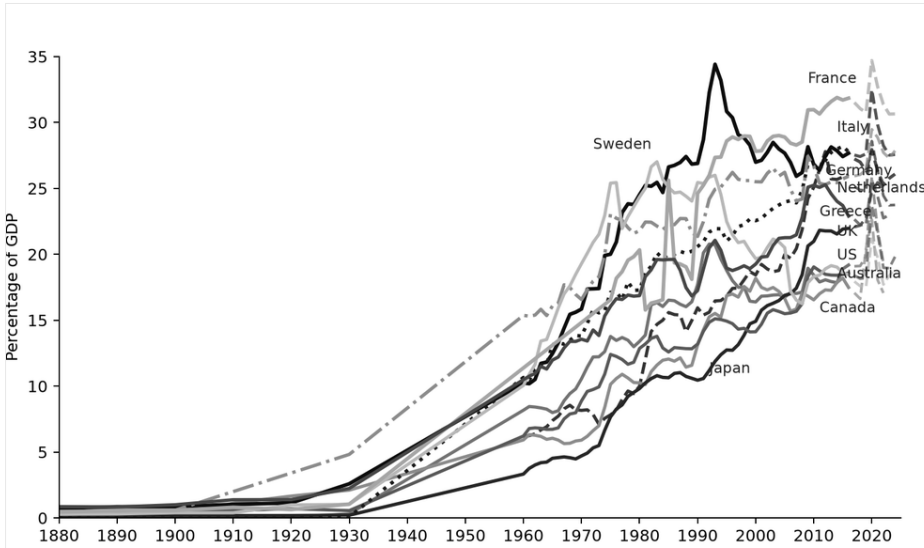


Figure 9-4: Social spending, OECD countries, 1880–2016

Source: *Our World in Data*, Ortiz-Ospina & Roser 2016b, based on data from Lindert 2004 and OECD 1985, 2014, 2017. The Organisation for Economic Co-operation and Development includes thirty-five democratic states with market economies.

Recreated/status



Solid segments show book period through 2016; dashed segments use current OWID/OECD successor data after 2016.

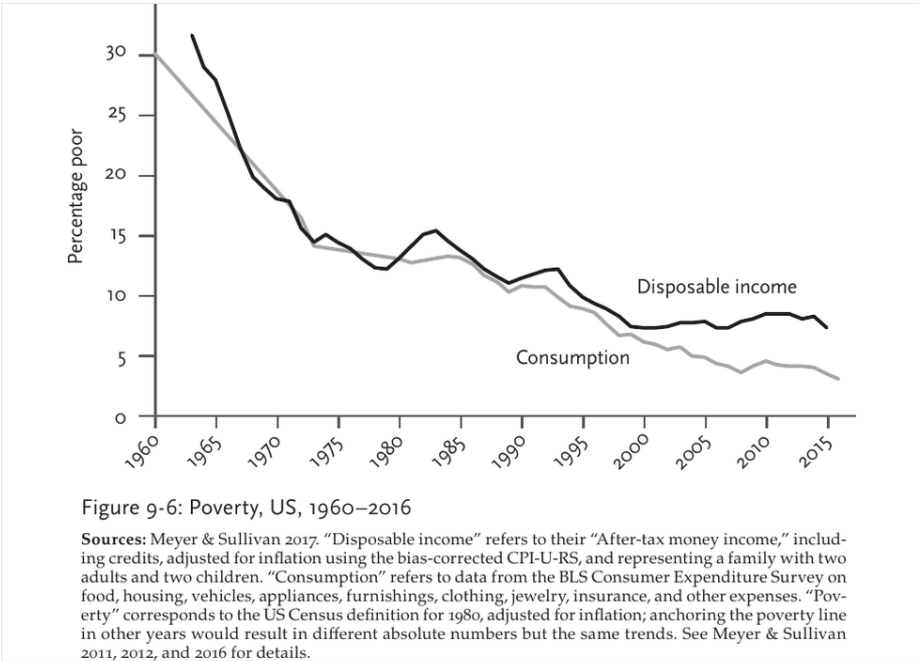
17. Figure 9-6: Poverty, US, 1960-2016

Status: verified_reproduction | Selected: figure_9_6_extended_comparison.png | Source: base_track_d_current

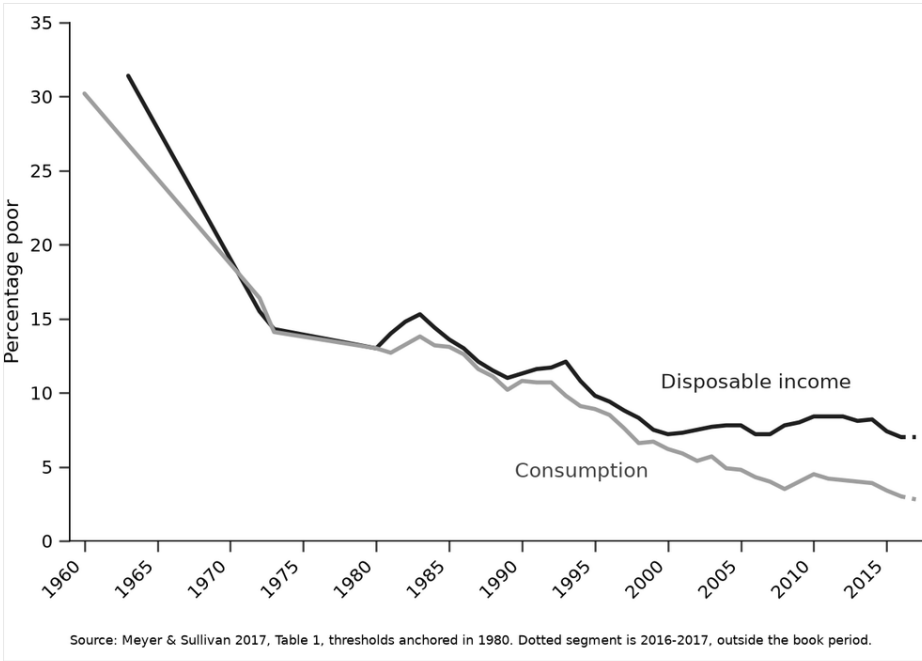
The cited report was recovered and Table 1 contains the two plotted book-period series. Values are extracted from the source table, not digitized from Pinker's figure.

Figure 9-6 extended comparison

PDF chart reference



Recreated

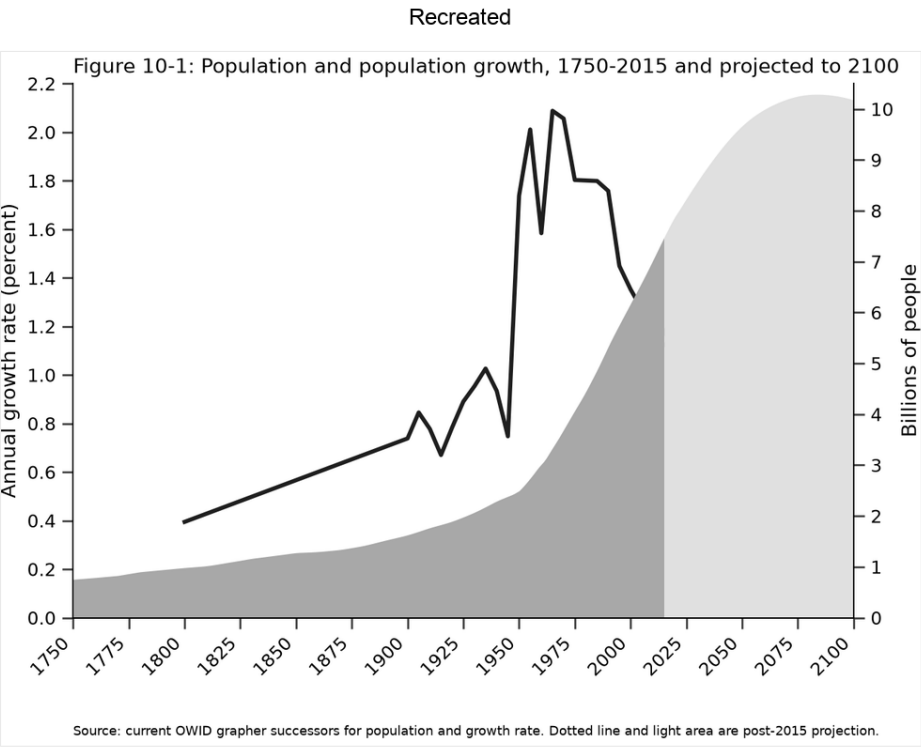
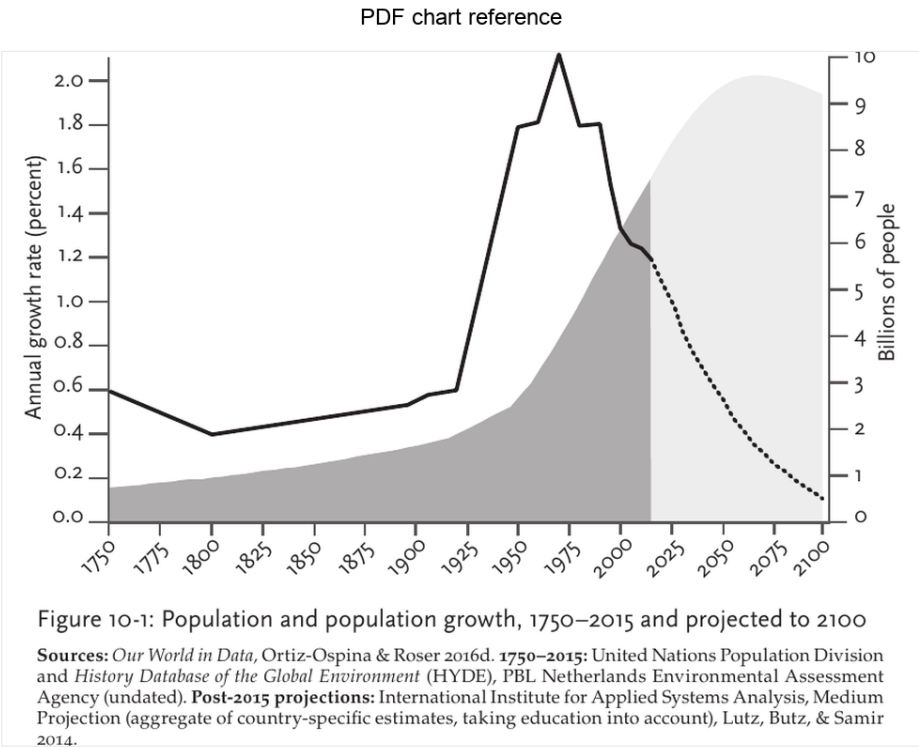


18. Figure 10-1: Population and population growth, 1750-2015 and projected to 2100

Status: updated_equivalent | Selected: figure_10_1_extended_comparison.png | Source: base_track_d_current

Uses current OWID grapher successor data (HYDE 2023, Gapminder 2022, UN WPP 2024) rather than the exact 2016 OWID/HYDE/IIASA source vintage cited by the book.

Figure 10-1 extended comparison



19. Figure 10-2: Sustainability, 1955-2109

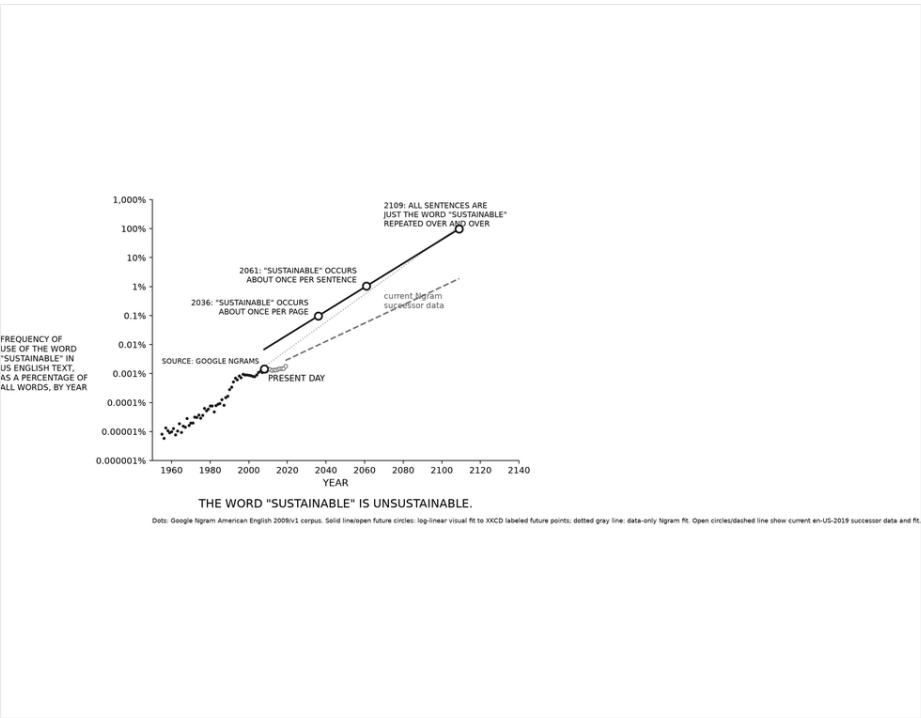
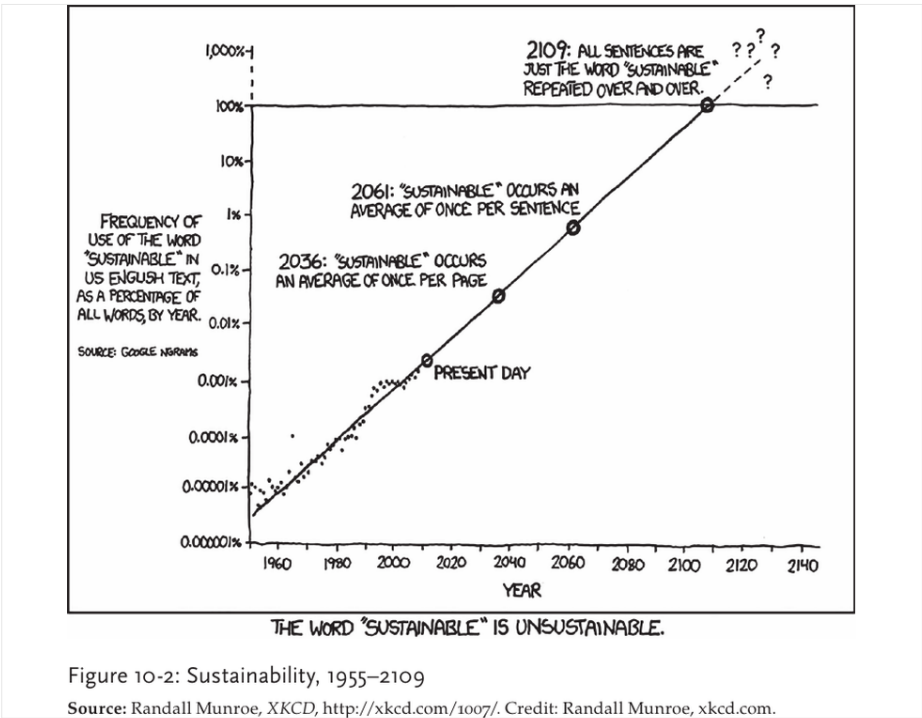
Status: updated_equivalent | Selected: figure_10_2_extended_comparison.png | Source: base_track_d_current

Updated-equivalent is retained because the official XKCD image, Supplemental PDF crop, and Google Ngram source family are recovered. The solid recreated projection is calibrated to XKCD's labeled future points; the data-only Ngram fit is retained as a dotted diagnostic and does not match every label year. No values were digitized from Pinker's printed figure.

Figure 10-2 extended comparison

Supplemental PDF reference

Recreated from Ngram data



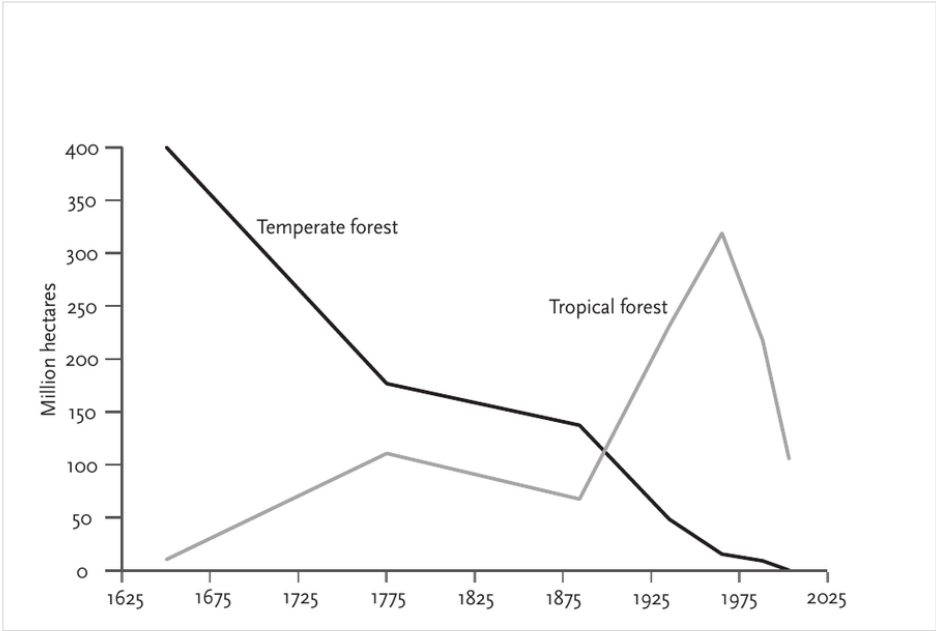
20. Figure 10-4: Deforestation, 1700-2010

Status: manual_review_needed | Selected: figure_10_4_extended_status_comparison.png | Source: base_track_d_current

The Supplemental PDF source line and FAO source graphic were recovered. The underlying Williams 2002 / FAO 2010b estimate table was not recovered, so no plotted reconstruction or extension was generated. Pinker's plotted values were not digitized.

Figure 10-4: Deforestation, 1700-2010

Supplemental Graphics PDF reference



Reconstruction status

Figure 10-4: no comparable extension

The book-period chart is already blocked on recovery of the Williams 2002 / FAO 2010b estimates used by FAO SOFO 2012 Figure 2.
Current forest-loss datasets such as FAO FRA 2025, OWID forest-area successors, and Global Forest Watch tree-cover loss are not methodologically comparable to the 1700-2010 historical deforestation estimates by tropical/temperate forest type.
No extension is plotted until the original source table is recovered or a clearly commensurable successor is identified.

No plotted values from Pinker's chart were digitized or used as source data.
The next recovery target is the Williams 2002 / FAO 2010b estimate table behind FAO SOFO 2012 Figure 2.

21. Figure 10-5: Oil spills, 1970-2016

Status: partial_match | Selected: figure_10_5_extended_comparison_captioned.png | Source: base_track_d_current

Figure 10-5 original reference

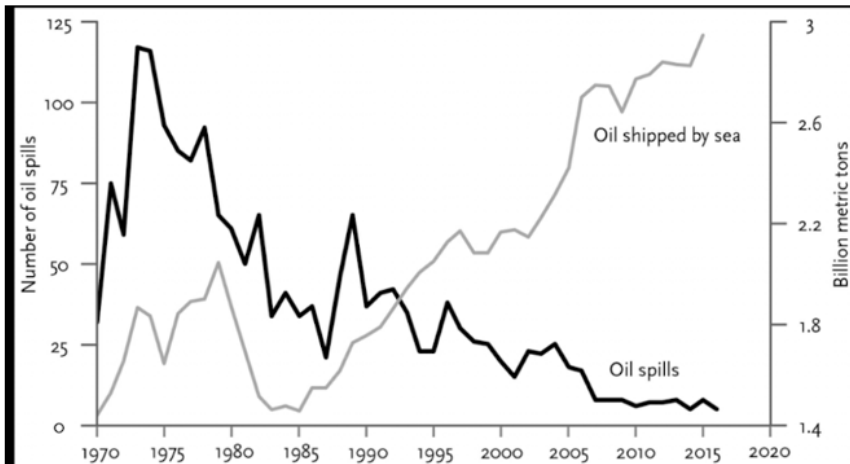
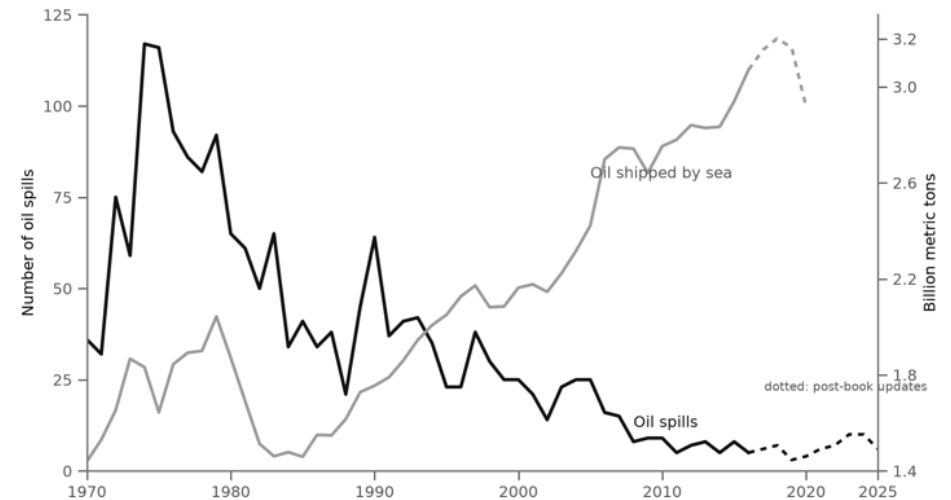


Figure 10-5: Oil spills, 1970-2016

Source: *Our World in Data*, Roser 2016r, based on data (updated) from the International Tanker Owners Pollution Federation, <http://www.itopf.com/knowledge-resources/data-statistics/statistics/>. Oil spills include all those that result in the loss of at least 7 metric tons of oil. Oil shipped consists of “total crude oil, petroleum product, and gas loaded.”

Extended reconstruction



Solid: book period through 2016. Dotted: post-book updates; tanker-trade mirror extends to 2020, spill counts to 2025.

Caption

Figure 10-5: Oil spills, 1970-2016, with post-publication updates. The solid black line shows oil spills of at least 7 tonnes during the book period; the solid gray line shows oil shipped by sea / tanker trade during the book period. Book-period spill data come from ITOPF via OWID/Roser, and tanker-trade data come from a recovered annual UNCTADStat-style mirror for World crude oil loaded plus other tanker trade loaded. Dotted segments show post-publication updates: spill counts extend through 2025 and tanker-trade evidence extends through 2020. The tanker-trade extension is continuous with the recovered annual mirror through 2020, but ITOPF's current source trail indicates UNCTADStat data are available through 2023; the accessible live UNCTAD API was not used for 2021-2023 because its overlap values do not match the book/RMT scale. Remaining uncertainty: the exact Roser 2016r/ITOPF data snapshot and a validated 2021-2023 same-scale UNCTAD export remain unresolved.

22. Figure 10-6: Protected areas, 1990-2014

Status: verified_reproduction | Selected: figure_10_6_extended_comparison_captioned.png | Source: base_track_d_current

Figure 10-6 original reference

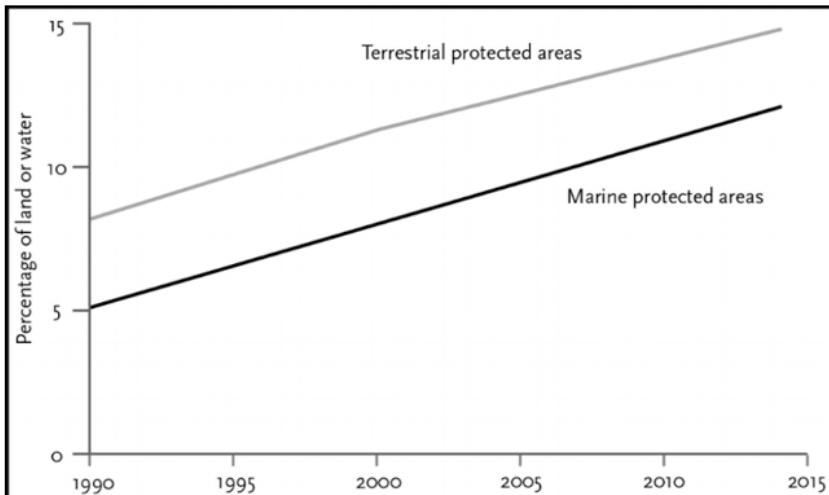
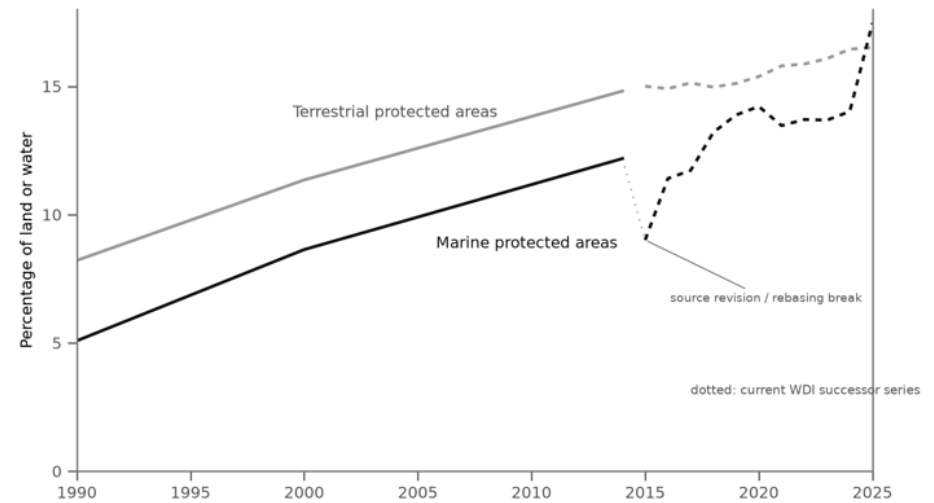


Figure 10-6: Protected areas, 1990-2014

Source: World Bank 2016h and 2017, based on data from the United Nations Environment Programme and the World Conservation Monitoring Centre, compiled by the World Resources Institute.

Extended reconstruction



Solid: archived WDI 2017 book period. Dotted: current WDI successor values; marine break reflects source revision/rebasing, not plotting error.

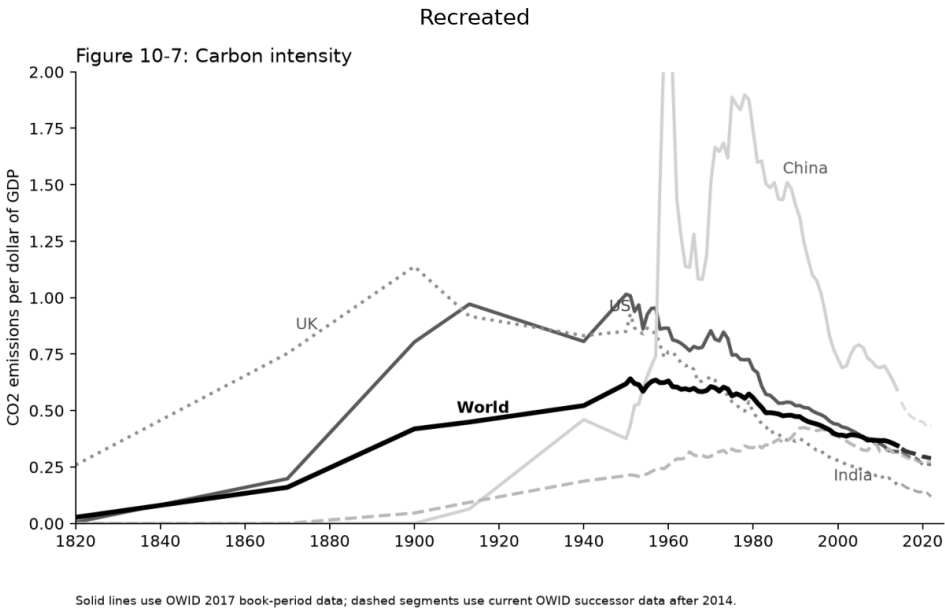
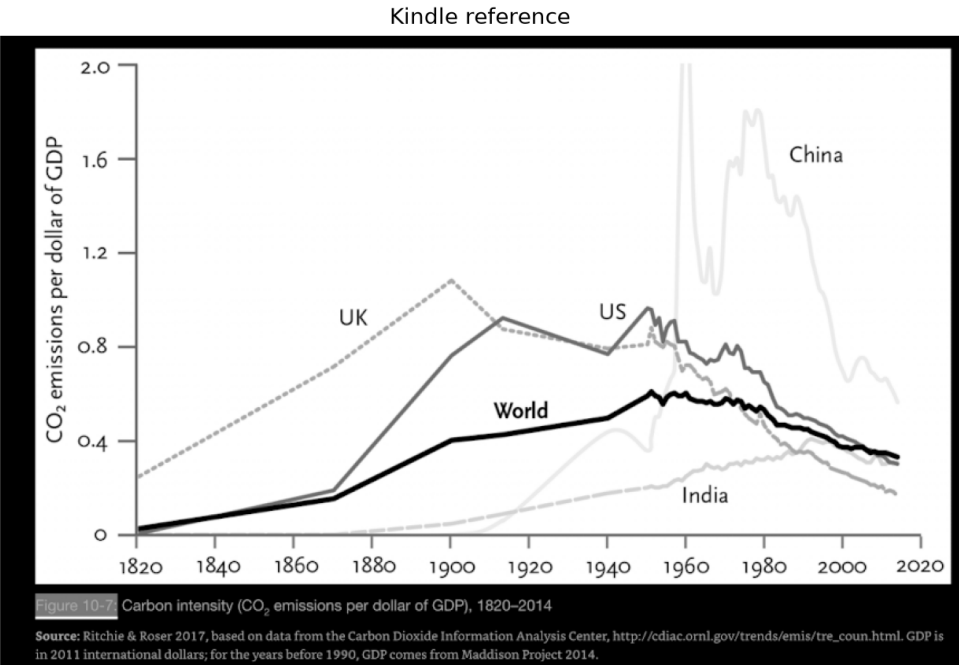
Caption

Figure 10-6: Protected areas, 1990-2014, with post-publication updates. Solid lines reproduce the book period from the archived World Bank WDI bulk ZIP captured by the Internet Archive on 2017-10-12; gray is terrestrial protected areas and black is marine protected areas. Dotted lines show current World Bank WDI successor values after the book period, based on newer Protected Planet / WDPA / WD-OECM source metadata. The dotted extension is not continuous with the archived book-period source: the marine series visibly jumps downward because archived WDI reports 12.18% for 2014, while current WDI reports 8.30% for 2014 and 9.01% for 2015. This discontinuity reflects source revision or rebasing, not a plotting error. Remaining uncertainty: a post-2014 release that is methodologically continuous with the archived 2017 WDI book-period values has not been recovered.

23. Figure 10-7: Carbon intensity (CO2 emissions per dollar of GDP), 1820-2014

Status: verified_reproduction | Selected: figure_10_7_extended_comparison.png | Source: base_track_d_current

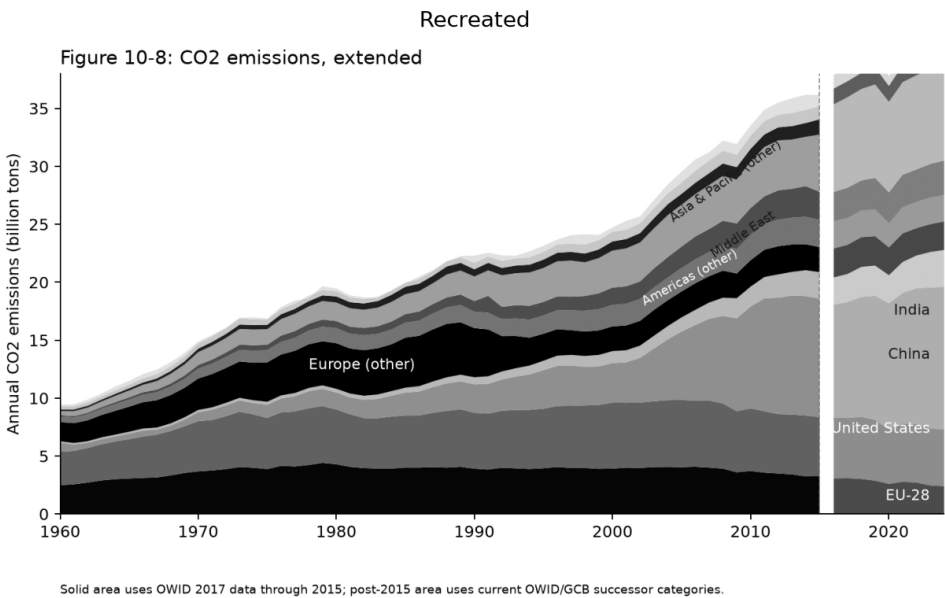
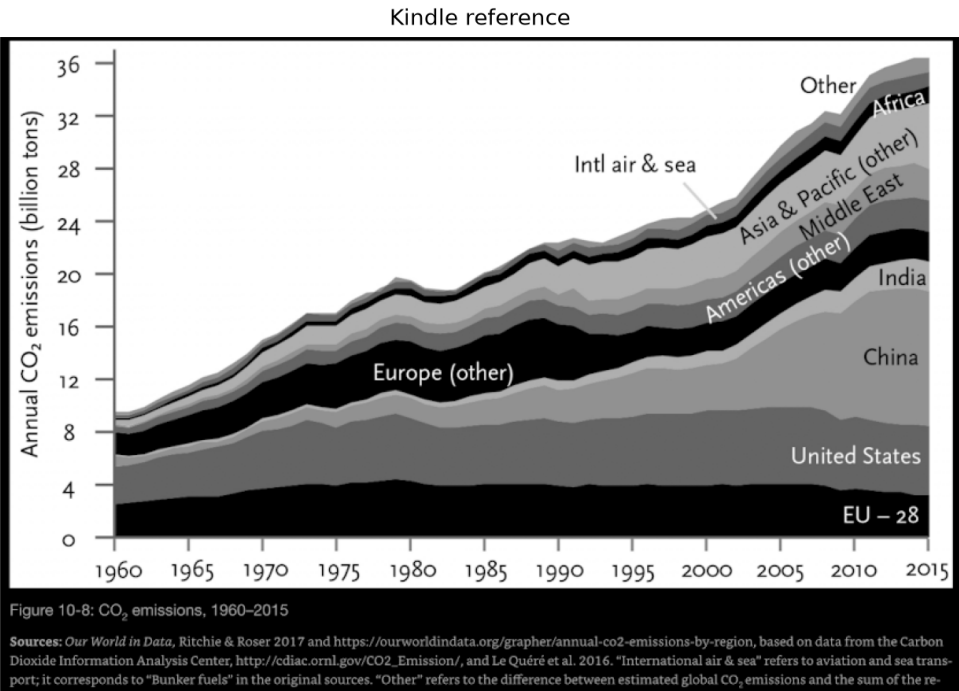
Figure 10-7 extended comparison



24. Figure 10-8: CO2 emissions, 1960-2015

Status: verified_reproduction | Selected: figure_10_8_extended_comparison.png | Source: base_track_d_current

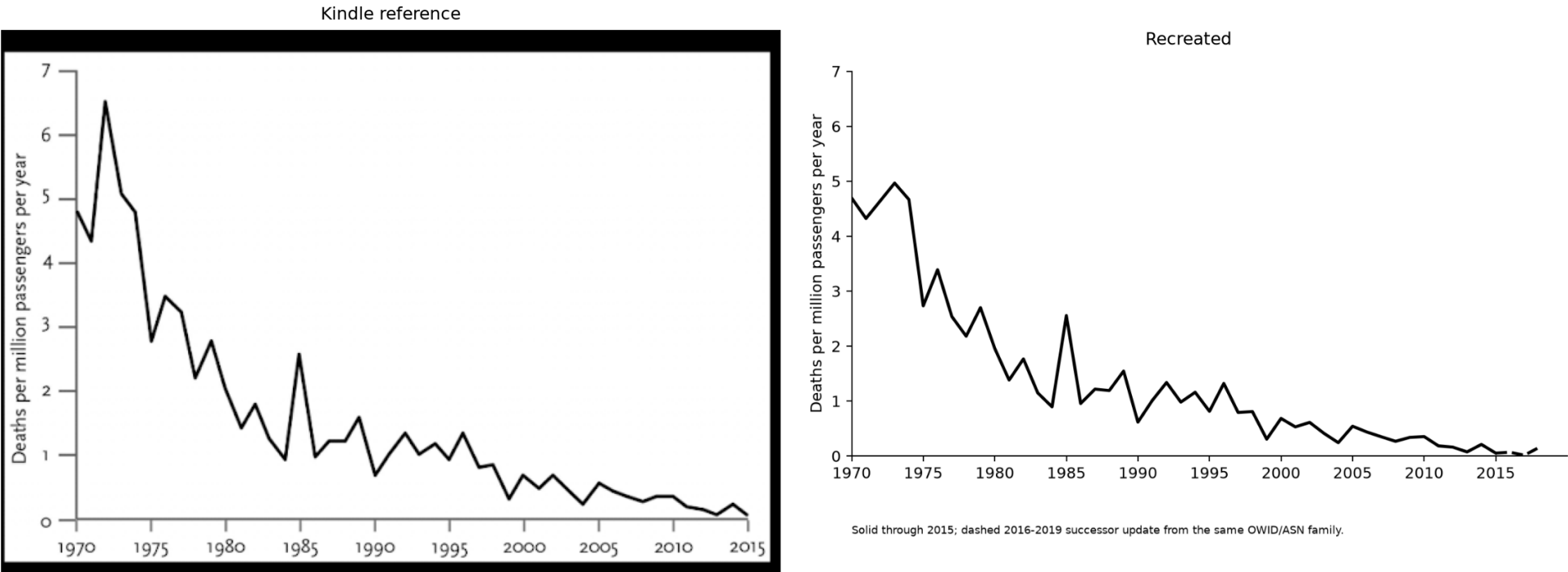
Figure 10-8 extended comparison



25. Figure 12-5: Plane crash deaths, 1970-2015

Status: partial_match | Selected: figure_12_5_extended_comparison.png | Source: base_track_d_current

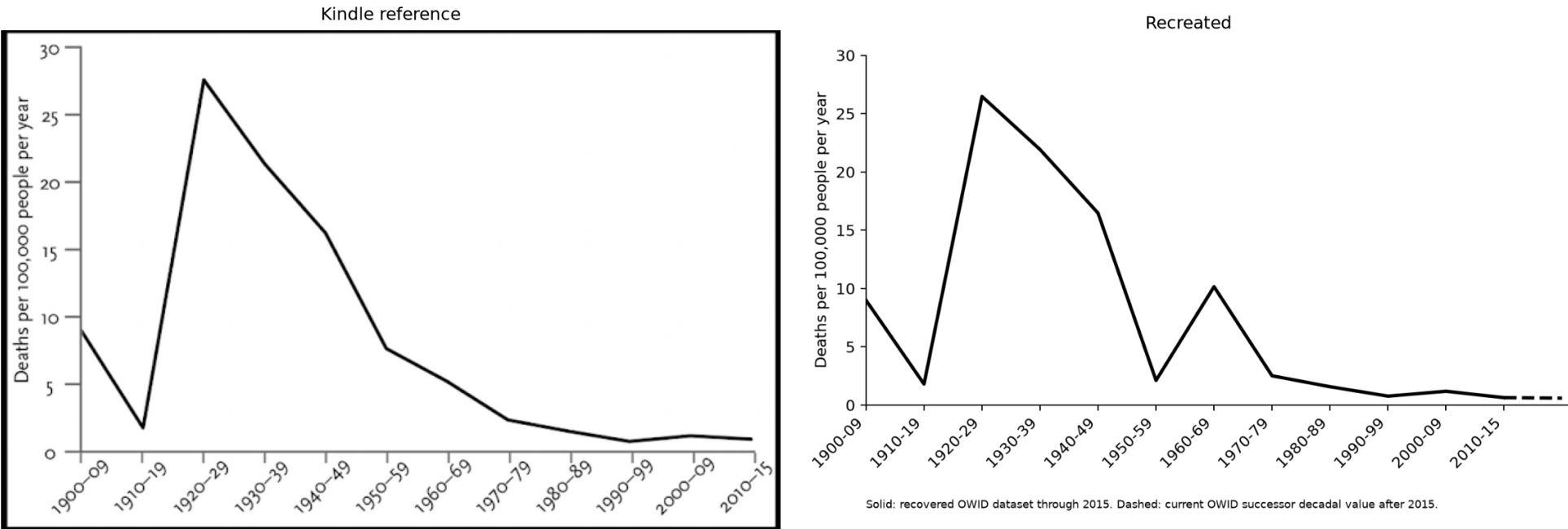
Figure 12-5 extended comparison



26. Figure 12-8: Natural disaster deaths, 1900-2015

Status: updated_equivalent | Selected: figure_12_8_extended_comparison.png | Source: base_track_d_current

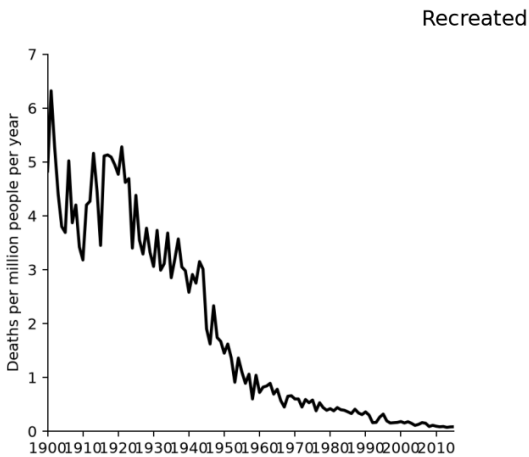
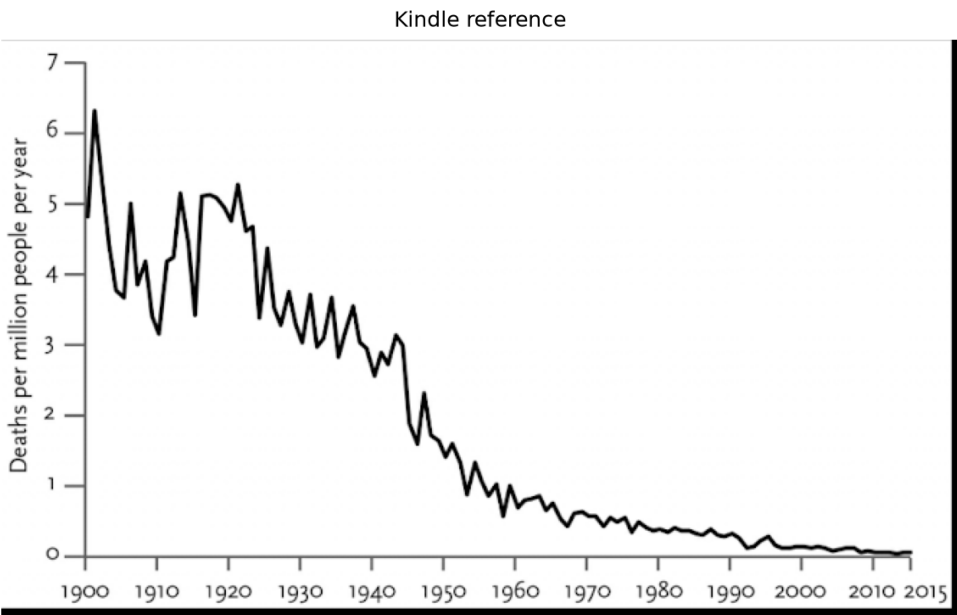
Figure 12-8 extended comparison



27. Figure 12-9: Lightning strike deaths, US, 1900-2015

Status: verified_reproduction | Selected: figure_12_9_extended_comparison.png | Source: base_track_d_current

Figure 12-9 extended comparison



Source: OWID/Roser 2016q-style NOAA and Lopez & Holle lightning fatality-rate dataset. No methodologically identical post-2015 extension was recovered in this pass.

28. Figure 19-1: Nuclear weapons, 1945-2015

Status: partial_match | Selected: figure_19_1_extended_comparison.png | Source: base_track_d_current

Actual Kindle chart-page capture is now present. The reconstruction remains partial because it uses a current OWID successor line series rather than the cited HumanProgress/FAS 2016 table and does not reproduce the book's stacked-area presentation.

Figure 19-1 extended comparison

